4.8 低维系统中的电子态与表面电子态

4.8 低维系统中的电子态

What does low-D mean: Characterized Length is comparable with or less than the de Broglie wavelength of carriers at Fermi level

Why low-D semiconductors are more interesting than bulk?

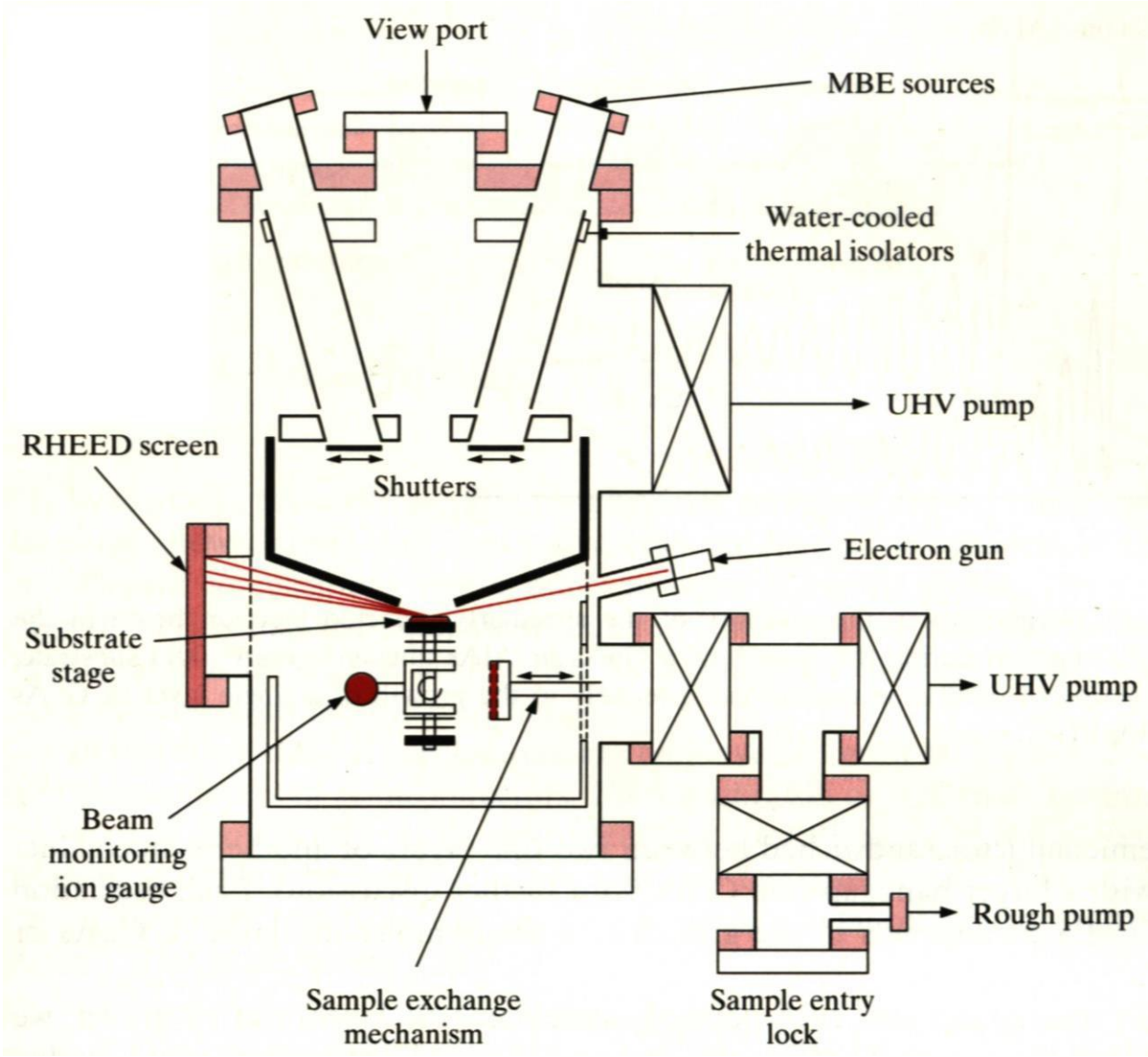
–**New or better devices:** better lasers (production of Quantum Dot lasers), single electron transistors, higher density memory, quantum computers etc

–**New physics**: new phenomena such as Fractional QHE, Luttinger Liquid etc

分子束外延设备

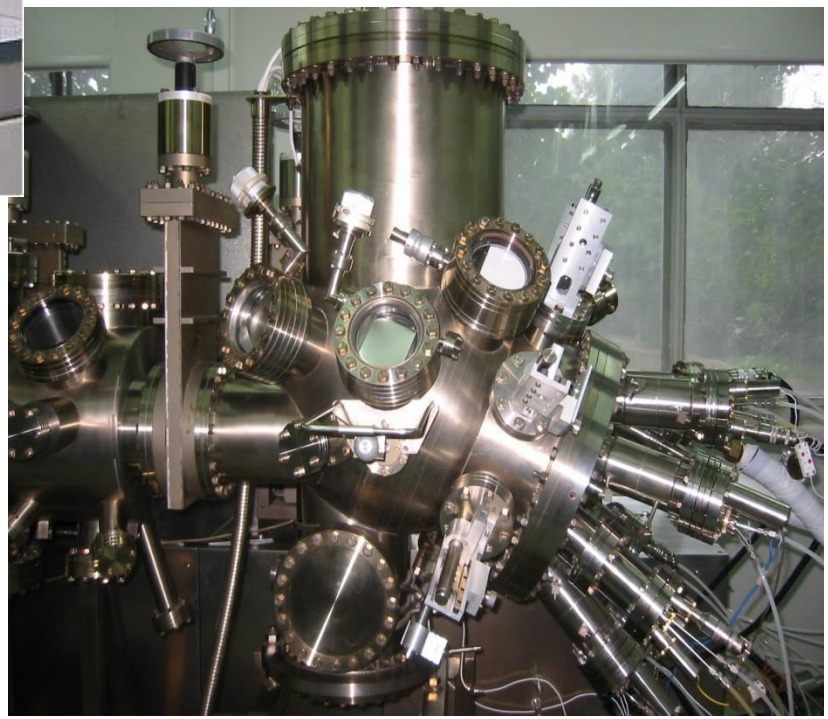
MBE
(Molecular Beam Epitaxy)

MBE是一种在超高真空环境下精确控制薄膜生长的技术，广泛用于制备高质量半导体薄膜等。



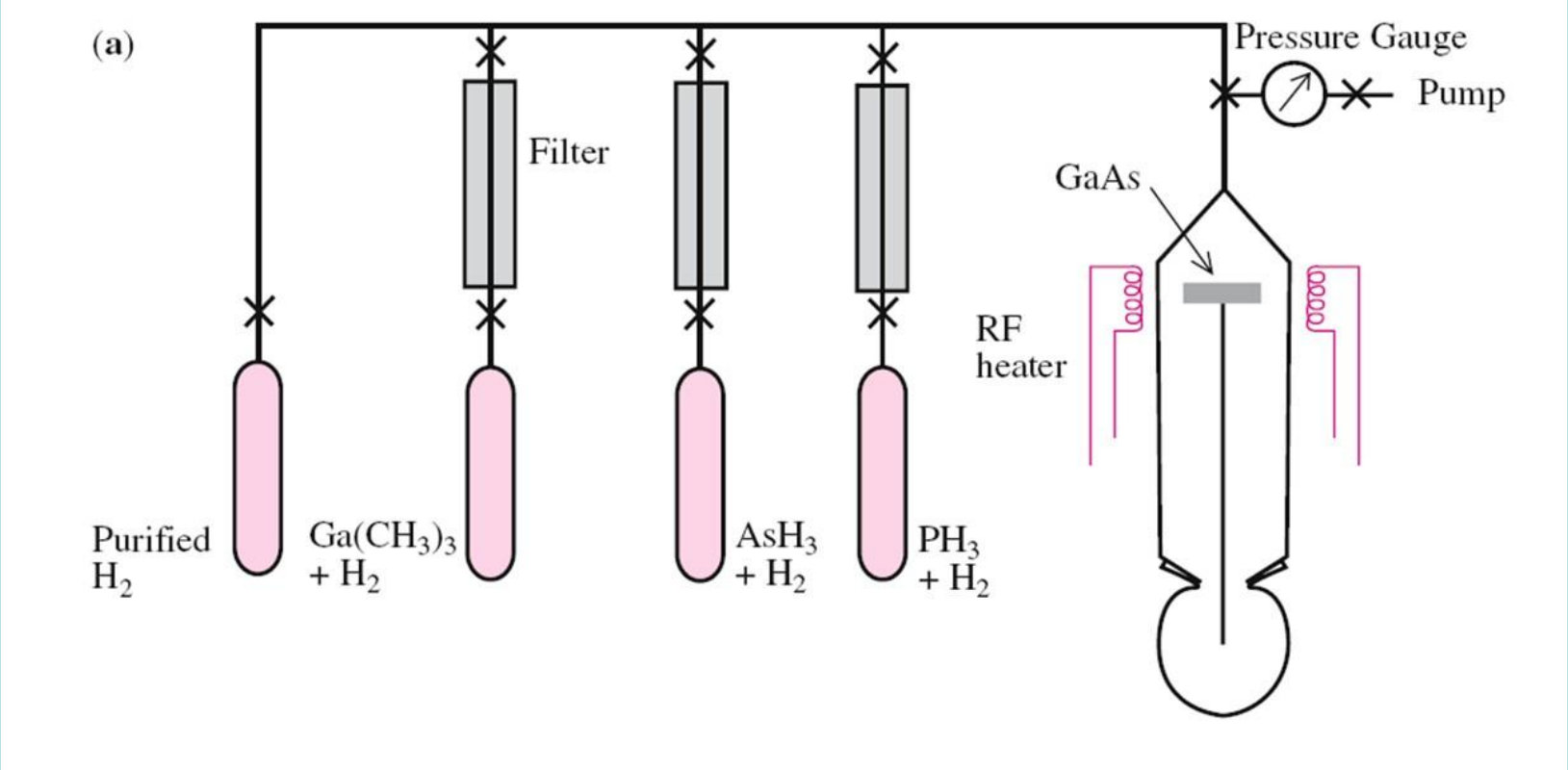


III-V族MBE



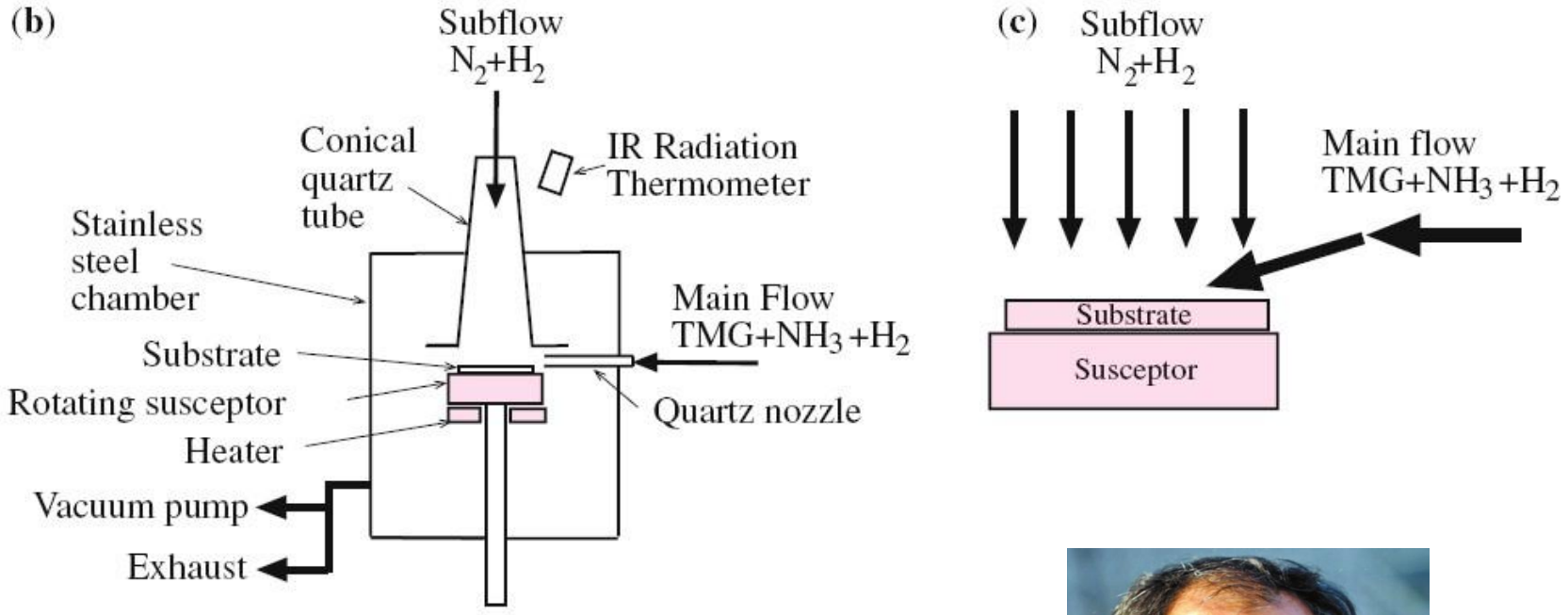
II-VI族MBE
(香港科大)

Metal-organic Chemical Vapor Deposition (MOCVD)



$$Ga(CH_3)_3 + AsH_3 \rightarrow 3CH_4\uparrow + GaAs.$$

Two-flow MOCVD machine by Nakamura et al for growing GaN



Shuji Nakamura(中村修二)



"for the invention of efficient blue light-emitting diodes which has enabled bright and energy-saving white light sources"

因发明高效蓝光二极管获诺贝尔奖，这一成果实现了明亮且节能的白光源。

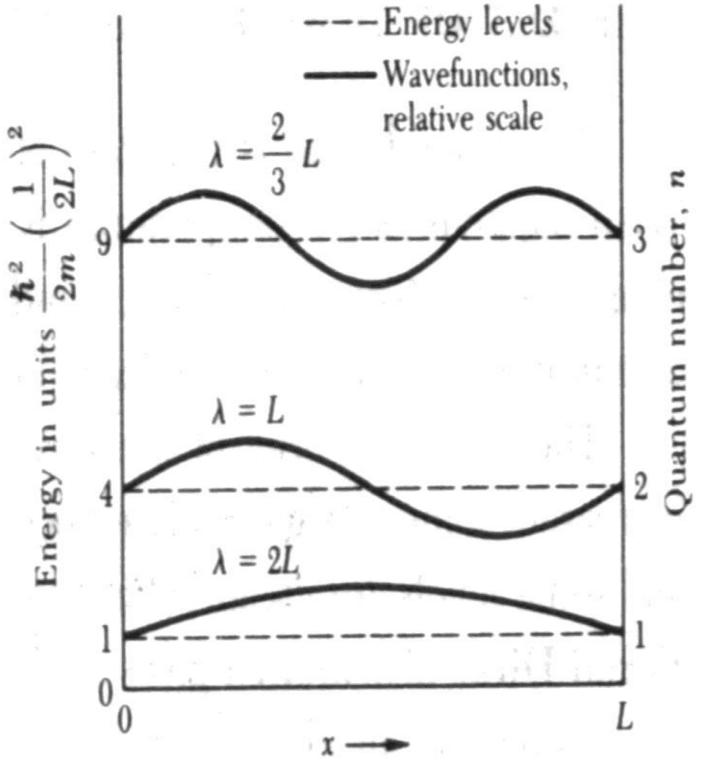
无限深量子阱：阱宽 L

$$\psi_n = A\varphi_n(z)\exp\left(ik_x x + ik_y y\right)$$

$$\varphi_n(0) = \varphi_n(L) = 0$$

$$\hat{H}\psi_n = -\frac{\hbar^2}{2m}\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}\right)\psi_n$$

$$= \frac{\hbar^2}{2m}\left[k_x^2 + k_y^2 + \left(\frac{n\pi}{L}\right)^2\right]\psi_n$$



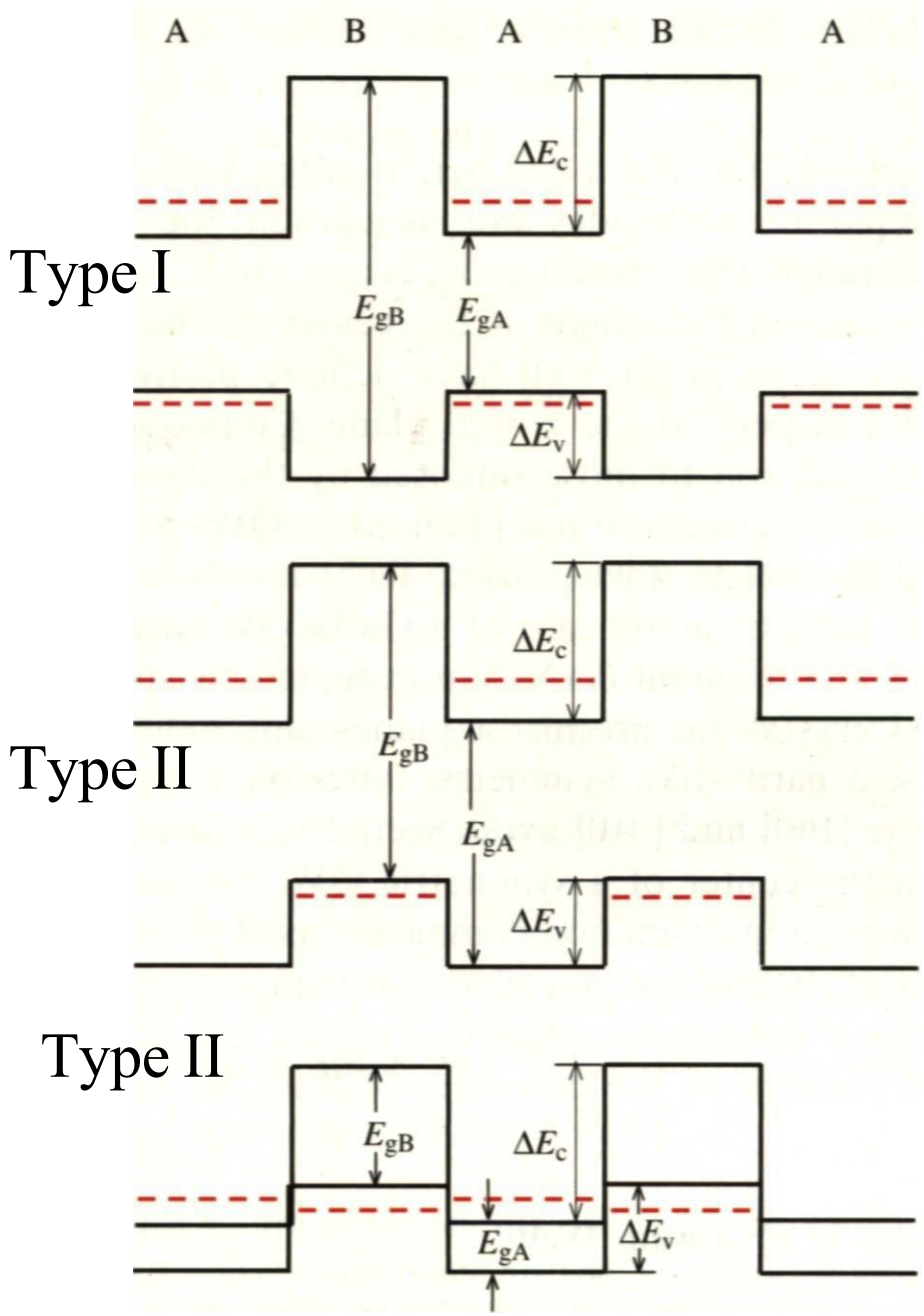
$$\varphi_n(z) = \sin\left(\frac{n\pi z}{L}\right) \qquad E_n(k) = \frac{\hbar^2}{2m}\left(k_x^2 + k_y^2\right) + \frac{\hbar^2}{2m}\left(\frac{n\pi}{L}\right)^2$$

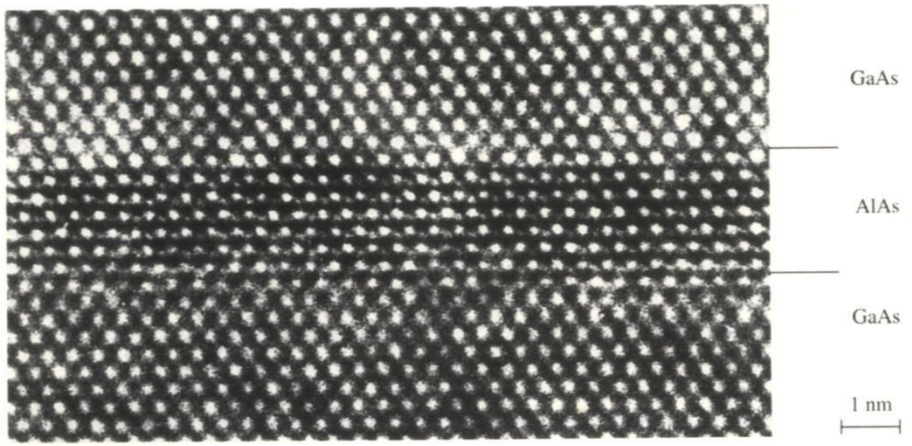
一个子带(subband)的态密度：

$$\frac{N}{S} = \frac{m}{\pi\hbar^2}\sum_n\left[E_F - \frac{\hbar^2}{2m}\left(\frac{n\pi}{L}\right)^2\right]\Theta\left[E_F - \frac{\hbar^2}{2m}\left(\frac{n\pi}{L}\right)^2\right]$$

超晶格(superlattice)

由两种或多种不同材料的薄层交替生长形成的人工周期性结构。



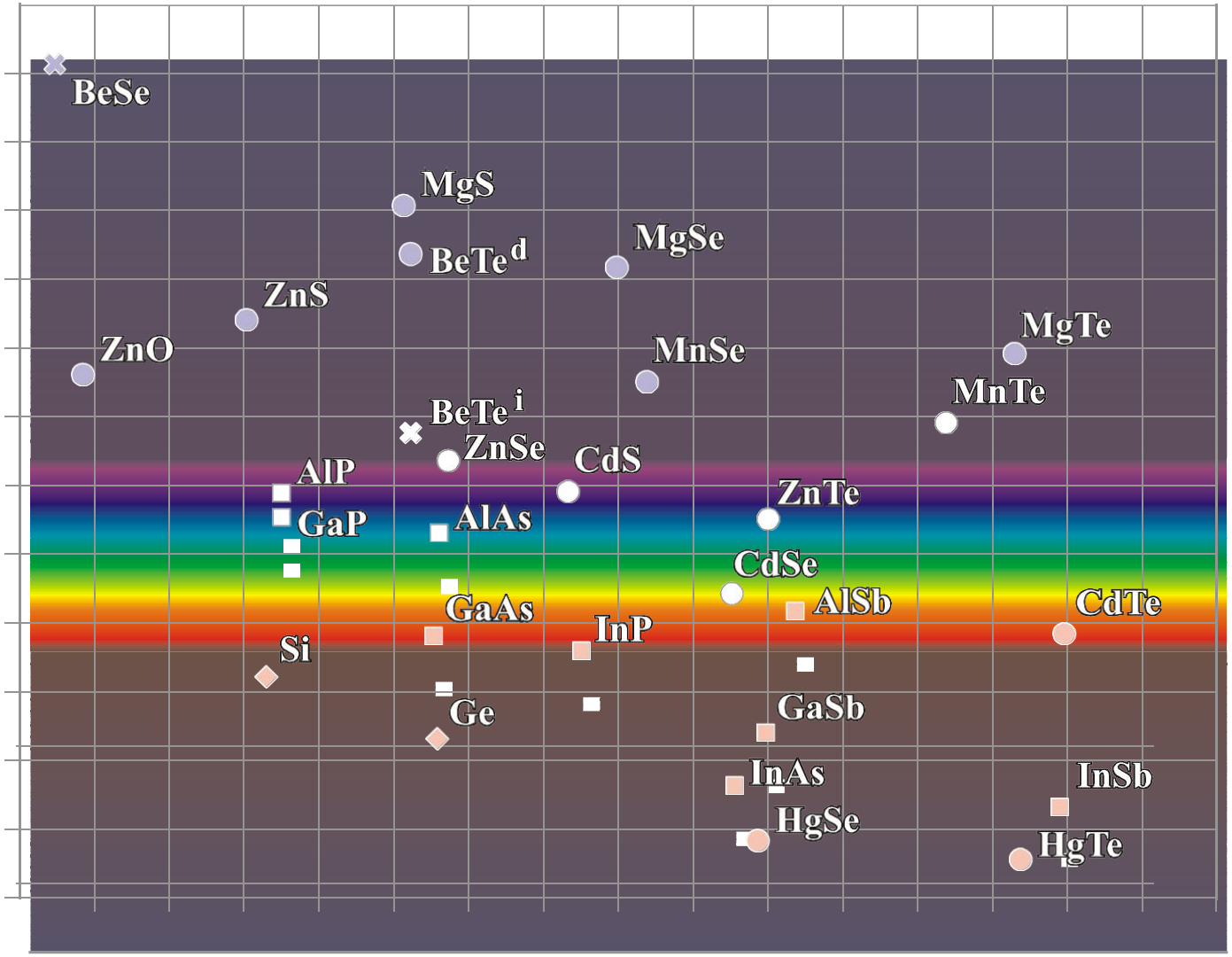


GaAs/AlAs 超晶格，可清晰看到交替排列的 GaAs 和 AlAs 层。

I：窄禁带材料（如 A）的导带和价带完全位于宽禁带材料（如 B）的带隙内（*EgB>EgA*）

II：能带呈错开排列。电子和空穴分别被限制在不同材料层中。

多种半导体材料



中间的彩色条带可能代表某一关键物理性质的渐变（如禁带宽度、电子亲和能等），不同颜色对应性质的不同数值范围。

- **Type III quantum wells work. HgTe has a negative band gap!**
- **Tuning the thickness of the HgTe/CdTe quantum well leads to a topological quantum phase transition into the QSH state.**

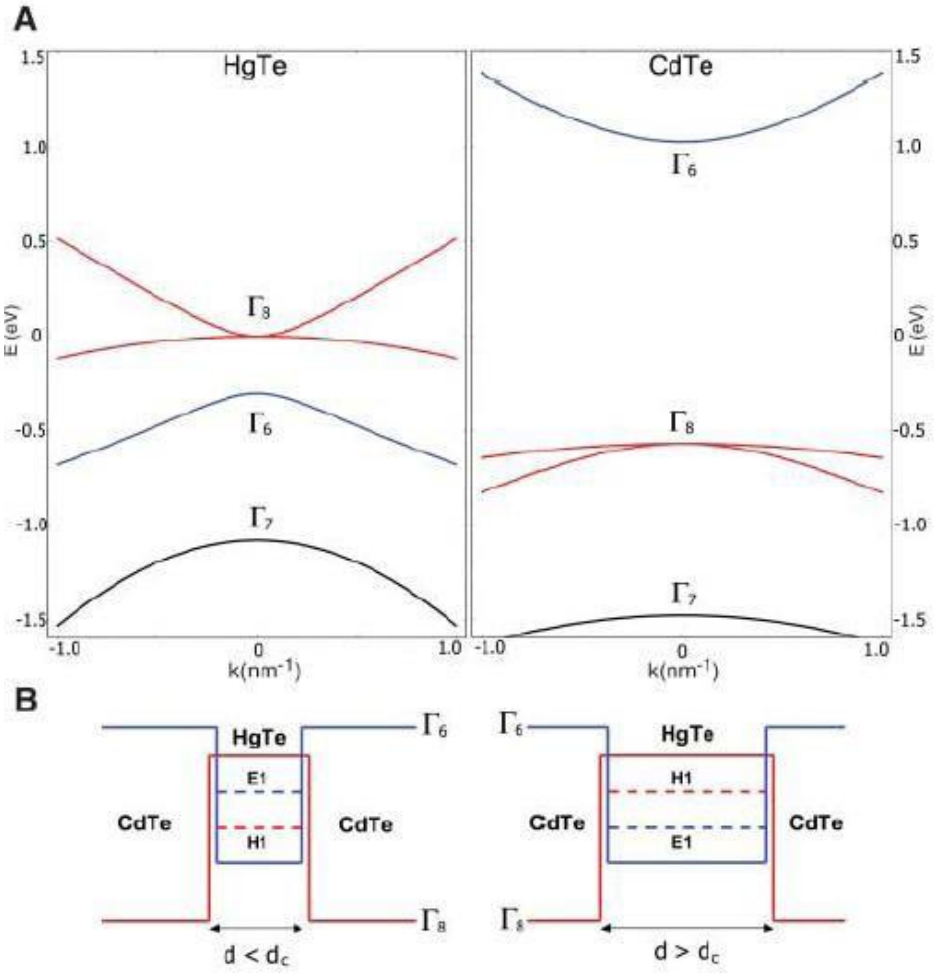
$$\Psi = \left(\left|\Gamma_6, \frac{1}{2}\right\rangle, \left|\Gamma_6, -\frac{1}{2}\right\rangle, \left|\Gamma_8, \frac{3}{2}\right\rangle, \left|\Gamma_8, \frac{1}{2}\right\rangle, \left|\Gamma_8, -\frac{1}{2}\right\rangle, \left|\Gamma_8, -\frac{3}{2}\right\rangle \right)$$

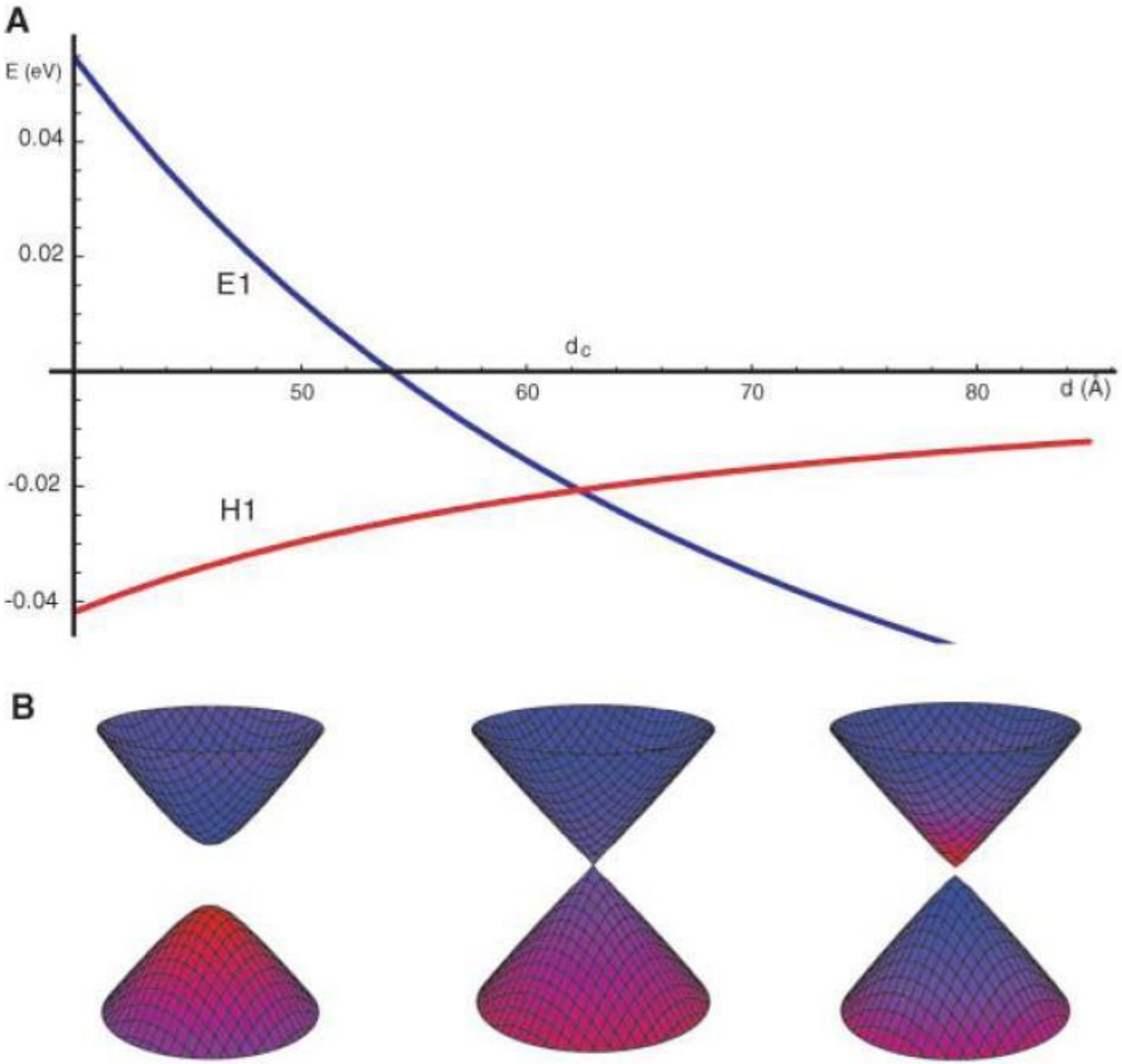
Start: Kane model

$$\mathcal{H}(\vec{k}) = \begin{pmatrix} E_c I_{2\times 2} + H_c & T_{2\times 4} \\ T^{\dagger}_{4\times 2} & E_v I_{4\times 4} + H_v \end{pmatrix}$$

$$H_c = \begin{pmatrix} \frac{\hbar^2 k^2}{2m^*} & 0 \\ 0 & \frac{\hbar^2 k^2}{2m^*} \end{pmatrix} \qquad T^{\dagger} = \begin{pmatrix} -\frac{1}{\sqrt{2}} P k_- & 0 \\ \sqrt{\frac{2}{3}} P k_z & -\frac{1}{\sqrt{6}} P k_- \\ \frac{1}{\sqrt{6}} P k_- & \sqrt{\frac{2}{3}} P k_z \\ 0 & \frac{1}{\sqrt{2}} P k_- \end{pmatrix}$$

$$H_v = -\frac{\hbar^2}{2m_0}\left(\gamma_1 + \frac{5}{2}\gamma_2\right)k^2 + \frac{\hbar^2}{m_0}\gamma_2 (\vec{k}\cdot\vec{S})^2$$



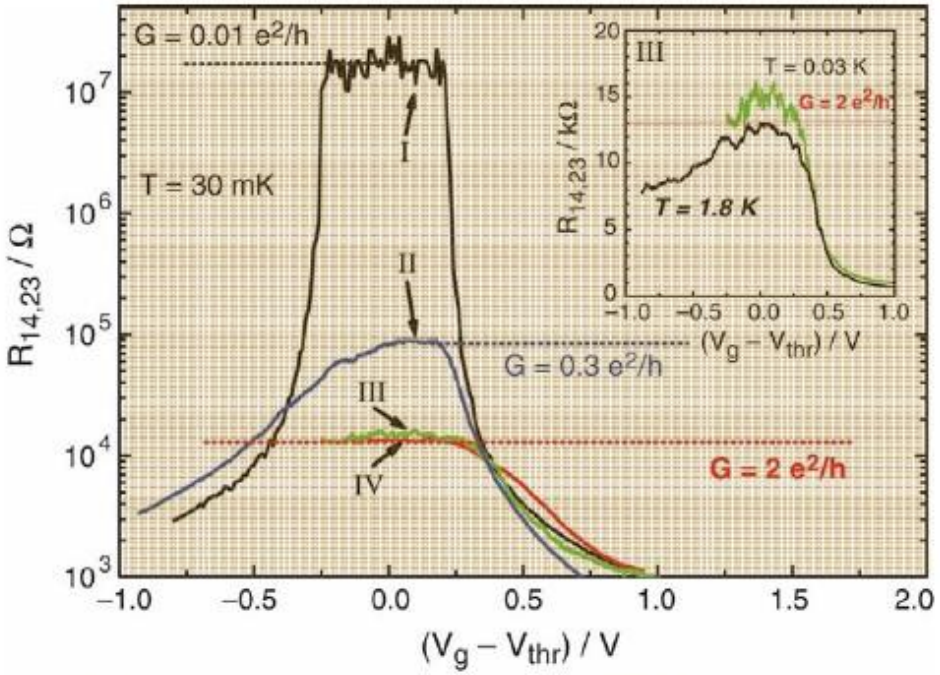


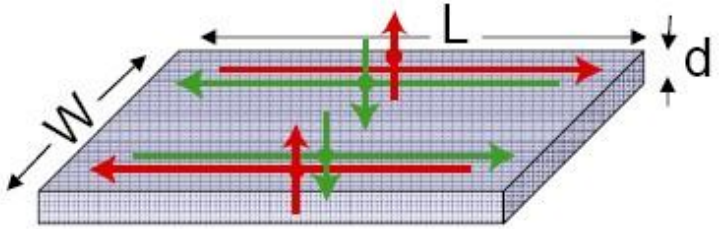
(Bernevig, Hughes and Zhang, Science 2006)

Observation of QSH insulator in HgTe quantum Wells

Konig, Wiedmann, Brune, Roth, Buhmann, Molenkamp, Qi, Zhang Science 2007

Measure 2 terminal conductance while tuning E_F through the bulk energy gap





I. d=5.5nm (normal) Insulating in gap

II-IV d=7.3nm (inverted) conducting in gap

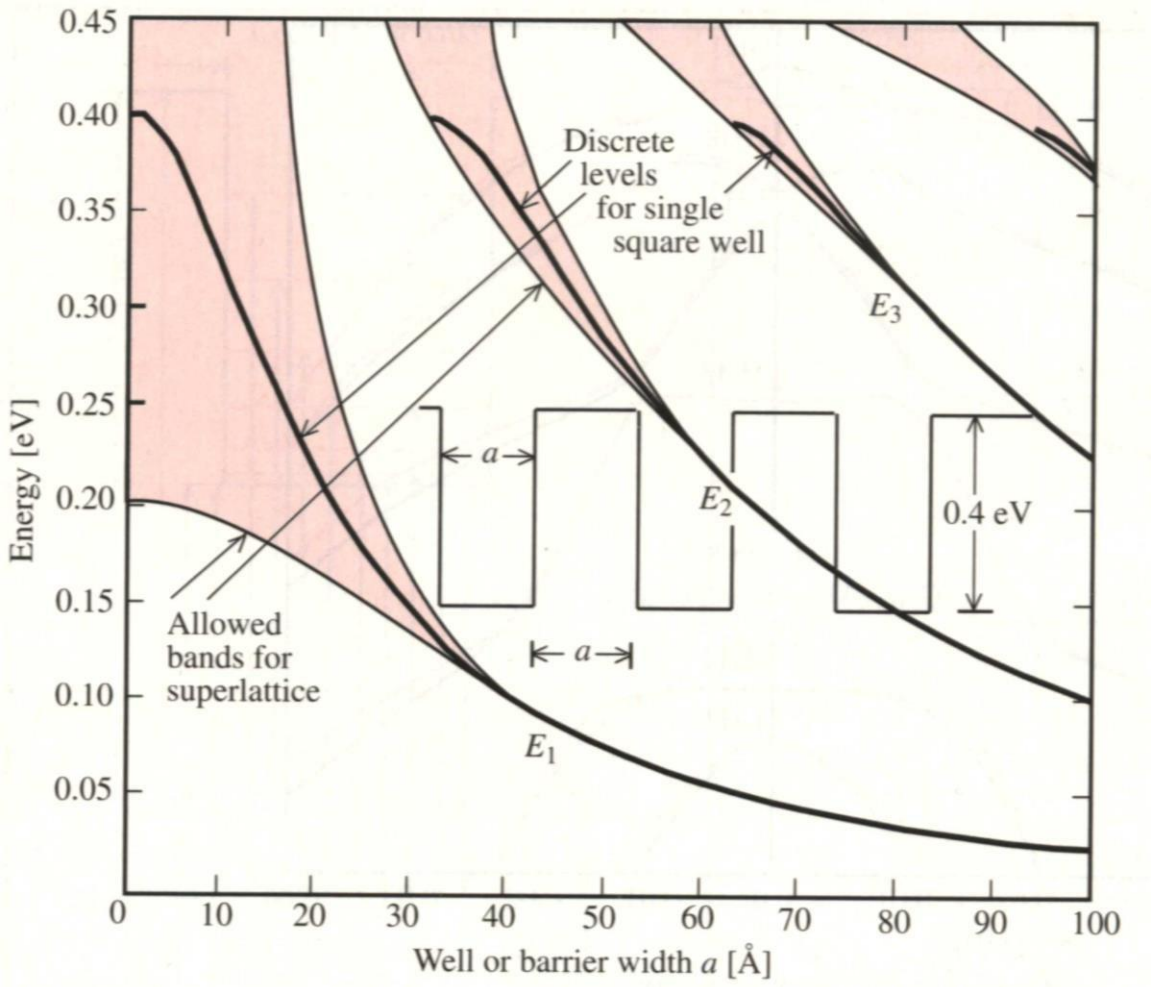
II. L = 20 μm (> L_{in})

III. L = 1 μm W = 1 μm

IV. L = 1 μm W = .5 μm

- Edge state conductance $2e^2/h$ observed independent of W for short samples ($L<L_{in}$)
- Bulk band inversion confirmed by Landau level evolution in magnetic field
- Magnetic field suppresses edge transmission

Kronig-Penney **模型**：研究周期性势场中电子行为的重要模型



Kronig - Penney模型通过分析周期性势场，揭示了晶体中电子能带的形成机制；超晶格的允许能带与单个方阱离散能级呈现不同变化趋势。

Current 2D library.

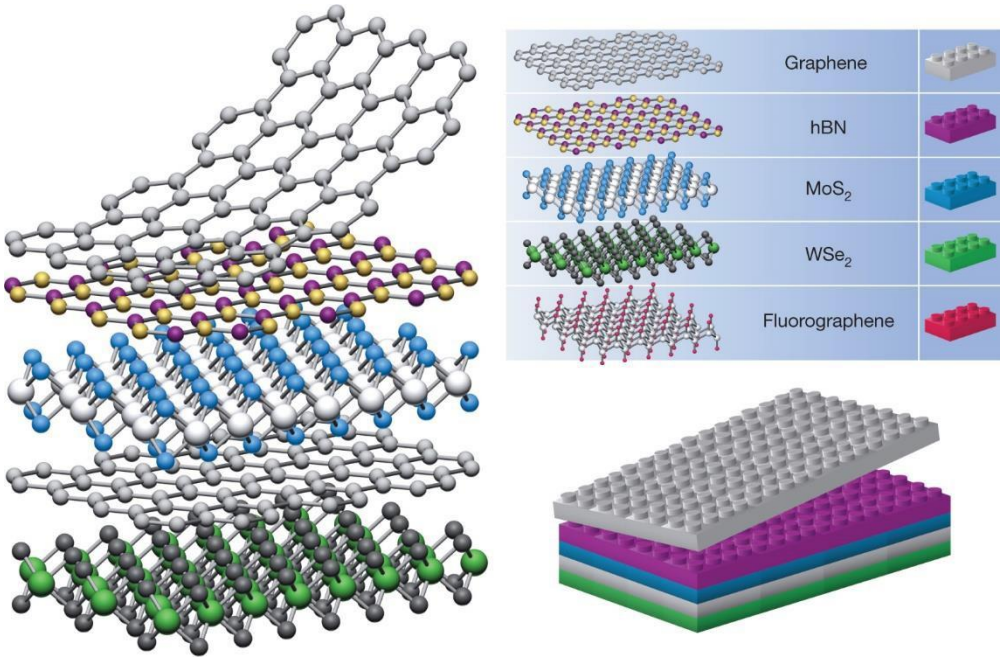
<table>
<tr><td>Graphene family</td><td>Graphene</td><td colspan="2">hBN
'white graphene'</td><td>BCN</td><td>Fluorographene</td><td>Graphene oxide</td></tr>
<tr><td rowspan="2">2D chalcogenIdes</td><td colspan="2" rowspan="2">MoS_2, WS_2, $MoSe_2$, WSe_2</td><td colspan="2" rowspan="2">Semiconducting dichalcogenides:
$MoTe_2$, WTe_2,
ZrS_2, $ZrSe_2$ and so on</td><td colspan="2">Metallic dichalcogenides:
$NbSe_2$, NbS_2, TaS_2, TiS_2, $NiSe_2$ and so on</td></tr>
<tr><td colspan="2">Layered semiconductors:
GaSe, GaTe, InSe, Bi_2Se_3 and so on</td></tr>
<tr><td rowspan="2">2D oxides</td><td>Micas, BSCCO</td><td colspan="2">MoO_3, WO_3</td><td rowspan="2" colspan="2">Perovskite-type:
$LaNb_2O_7$, $(Ca,Sr)_2Nb_3O_{10}$,
$Bi_4Ti_3O_{12}$, $Ca_2Ta_2TiO_{10}$ and so on</td><td>Hydroxides:
$Ni(OH)_2$, $Eu(OH)_2$ and so on</td></tr>
<tr><td>Layered Cu oxides</td><td colspan="2">TiO_2, MnO_2, V_2O_5,
TaO_3, RuO_2 and so on</td><td>Others</td></tr>
</table>

AK Geim & IV Grigorieva *Nature* **499**, 419-425 (2013)
doi:10.1038/nature12385

nature

范德华异质结构通过弱范德华力堆叠不同二维材料，形成新的人工结构。

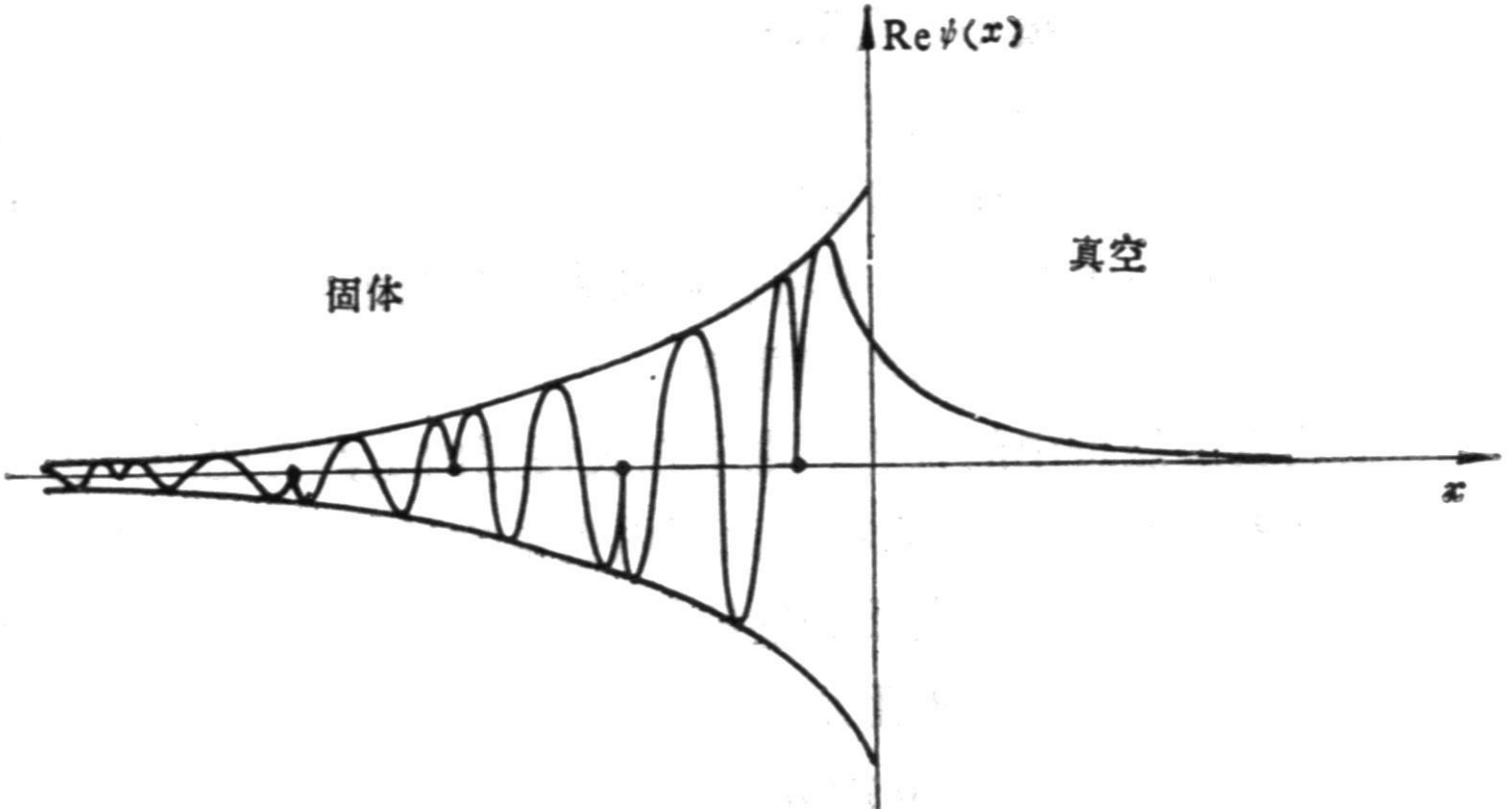
nature Building van der Waals heterostructures



AK Geim & IV Grigorieva
Nature **499**, 419-425 (2013)

范德华异质结构的构建，突破了传统晶体生长限制，可自由组合不同性质的二维材料（如导体、半导体、绝缘体），为调控电子态、探索新物理现象（如莫特绝缘态、量子霍尔效应等）及开发高性能器件（如柔性电子、高频晶体管）提供了广阔空间。

表面电子态



表面电子态是指局域在固体表面区域的电子状态。

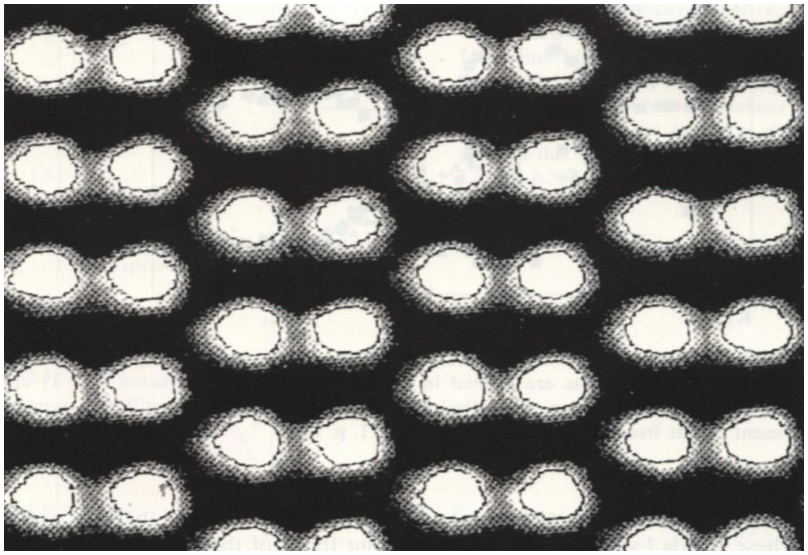
在固体中，波函数呈现出周期性振荡，反映了晶体内部周期性势场对电子的影响；而在真空区域，波函数迅速衰减。

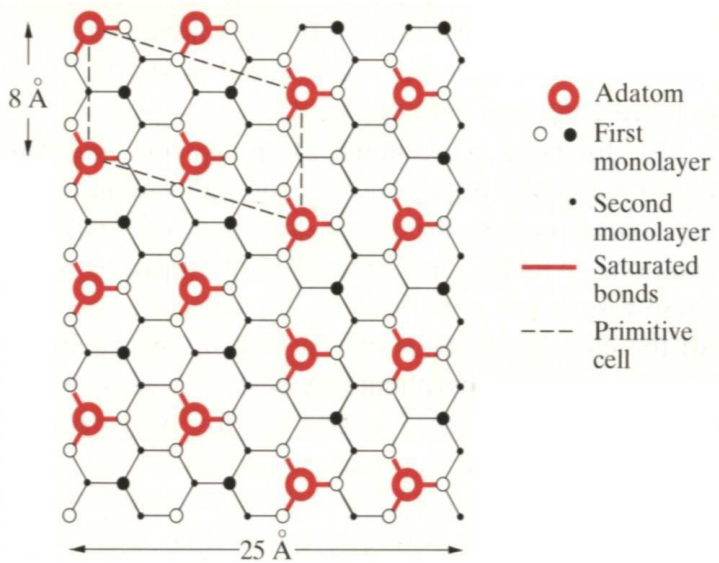
表面弛豫（relaxation）和重构(reconstruction)

表面弛豫：指晶体表面原子层的间距发生变化。

表面重构：表面原子重新排列，形成与体内不同的晶胞结构。

Ge(111)





通过Ge(111)的例子可以看到，表面原子通过弛豫和重构，形成了不同于体内的结构，这种变化显著影响材料的表面电子态和物理化学性质。

表面电子态

- 从理想模型表面出发，研究考察晶体表面对电子能量本征态的影响
- 假设晶体表面位于z = 0处，则有：

$$V(z) = \begin{cases} V(z) & z < 0 \quad \text{Crystal} \\ V_0 & z > 0 \quad \text{Vacuum} \end{cases}$$

- 电子在晶体内部的能量$E < V_0$
- 满足：

  在界面z = 0处电子波函数和一阶微分连续

一维晶体的波函数和能级

- 电子波动方程：

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dz^2}+V(z)\psi=E\psi$$

- $Z>0$电子波函数：

$$\psi=ae^{-\alpha z}$$

$$\alpha=\sqrt{\frac{2m}{\hbar^2}(V_0-E)}$$

- $Z<0$电子波函数：采用近自由电子近似

- 下面推导晶体内部的波函数以及能级：
- 对于近自由电子近似下，有：

1）零级近似下的电子能量和波函数

$$H_0 = -\frac{\hbar^2}{2m}\frac{d^2}{dx^2} + \bar{V}$$

- **零级近似的波动方程为**

$$-\frac{\hbar^2}{2m}\frac{d^2}{dx^2}\psi^0 + \bar{V}\psi^0 = E^0\psi^0$$

- **它的解便是恒定 $\bar{V}$ 中自由粒子的解**

$$\psi_k^0(x) = \frac{1}{\sqrt{L}}e^{ikx},\quad E_k^0 = \frac{\hbar^2 k^2}{2m} + \bar{V}$$

- **晶格长度 $L = Na$。**

2）微扰下电子的能量本征值哈密顿量：

$$\mathrm{H} = H_0 + H', H_0 = -\frac{\hbar^2}{2m}\frac{d^2}{dx^2} + \bar{V}, H' = \mathrm{V(x)} - \bar{V} = \Delta V$$

- 根据微扰理论，电子能量本征值

$$E_k = E_k^0 + E_k^{(1)} + E_k^{(2)} + \cdots$$

- 能量一级修正：

$$E_k^{(1)} = \langle k|H'|k\rangle = \langle k|\mathrm{V(x)} - \bar{V}|k\rangle$$
$$= \langle k|\mathrm{V(x)}k\rangle - \langle k|\bar{V}|k\rangle = \bar{V} - \bar{V} = 0$$

下面求解一维晶体的波函数和能级

- 在布里渊区边界，：$k=\frac{\pi}{a}+\varepsilon, k'=k-\frac{2\pi}{a}=-\frac{\pi}{a}+\varepsilon$
- 由简并微扰理论可得：

- 忽略所有其它掺入的状态，把波函数写成

$$\psi = a\psi_{k'}^{0} + b\psi_{k}^{0}$$

其中ψ_k^0和$\psi_{k'}^0$为前面求解过的电子零级波函数

- 电子波函数：$\psi_k(z) = ae^{i\left(\frac{\pi}{a}+\varepsilon\right)z} + be^{-i\left(\frac{\pi}{a}-\varepsilon\right)z}$
- 将波函数带入薛定谔方程，得：

$$H_0\psi(z) + H'\psi(z) = E\psi(z)$$

- **其中** $H_0 = -\frac{\hbar^2}{2m}\frac{d^2}{dz^2}, \quad H' = V(z) - \overline{V} = \Delta V$

- **并考虑到：**

$$\left[-\frac{\hbar^2}{2m}\frac{d^2}{dx^2} + \overline{V}\right]\psi_k^0 = E_k^0\psi_k^0$$

$$\left[-\frac{\hbar^2}{2m}\frac{d^2}{dx^2} + \overline{V}\right]\psi_{k'}^0 = E_{k'}^0\psi_{k'}^0$$

- **得：**

$$a\left(E_k^0 - E + \Delta V\right)\psi_k^0 + b\left(E_{k'}^0 - E + \Delta V\right)\psi_{k'}^0 = 0$$

- 上式分别乘以 ψ^{0*}_{k} 和 $\psi^{0*}_{k'}$并积分，得到

$$(E_k^0 - E)a + V_1 b = 0$$

$$V_1 a + (E_{k'}^0 - E)b = 0 \qquad (*)$$

- 其中用到：

$$\langle k|\Delta V|k\rangle = \langle k'|\Delta V|k'\rangle = 0$$

- 以及

$$V_1 = \int \psi_k^0 * V(z)\psi_{k'}^0 * dz$$

- 作为 a 和 b 的代数方程，有解条件为

$$\begin{vmatrix} E_k^0 - E & V_1 \\ V_1 & E_{k'}^0 - E \end{vmatrix} = 0$$

- 解得能量本征值：

$$E = \frac{\hbar^2}{2m}\left(\frac{\pi}{a} + \varepsilon\right)^2 + |V_1|\left(-v \pm \sqrt{v^2+1}\right) + \overline{V}$$

$$v = \frac{\hbar^2 \pi \varepsilon}{ma|V_1|}$$

- 将能量本征值带入（*），可得，电子波函数的组合系数a、b，进而得到电子波函数的表达式

$$\psi_k(z) = b\left[\exp\left(i\frac{\pi}{a}z\right) + \frac{|V_1|}{V}\left(-v \pm \sqrt{v^2+1}\right)\exp\left(-i\frac{\pi}{a}z\right)\right]\exp(i\varepsilon z)$$

- 对于ε为实数，$\psi_k(z)$就是晶体中电子波函数

$$\psi_k(z) = b\left[\exp\left(i\frac{\pi}{a}z\right) + \frac{|V_1|}{V}\left(-v \pm \sqrt{v^2+1}\right)\exp\left(-i\frac{\pi}{a}z\right)\right]\exp(i\varepsilon z)$$

- 同时，**在界面$z = 0$处电子波函数和一阶微分连续。**即晶体界面内的波函数和界面外的波函数$\psi = ae^{-\alpha z}$总是满足匹配

□ 半无限长晶体内部：

- 在$z<0$处，如果ε为复数
- 可令$\varepsilon=-iq$，q为正数
- 则可令 $$v=\frac{\hbar^2\pi\varepsilon}{ma|V_1|}=-i\frac{\hbar^2\pi q}{ma|V_1|}=-i\sin(2\delta)$$

- 则$z<0$处波函数：

$$\psi_k(z)=c\left\{\exp\left[i\left(\frac{\pi}{a}z\pm\delta\right)\right]+\left|\frac{V_1}{V}\right|\exp\left[-i\left(\frac{\pi}{a}z\pm\delta\right)\right]\right\}\exp(qz)$$

- 则$z<0$处能量：

$$E=\frac{\hbar^2}{2m}\left[\left(\frac{\pi}{a}\right)^2-q^2\right]\pm|V_1|\left[1-\left(\frac{\hbar^2\pi q}{ma|V_1|}\right)^2\right]^{1/2}+\overline{V}$$

半无限长晶体内部：

- 根据界面z = 0处电子波函数和一阶微分连续条件，
- $z < 0$处波函数：

$$\psi_k(z) = c\left\{\exp\left[i\left(\frac{\pi}{a}z \pm \delta\right)\right] + \left|\frac{V_1}{V}\right|\exp\left[-i\left(\frac{\pi}{a}z \pm \delta\right)\right]\right\}\exp(qz)$$

- $z > 0$处波函数：

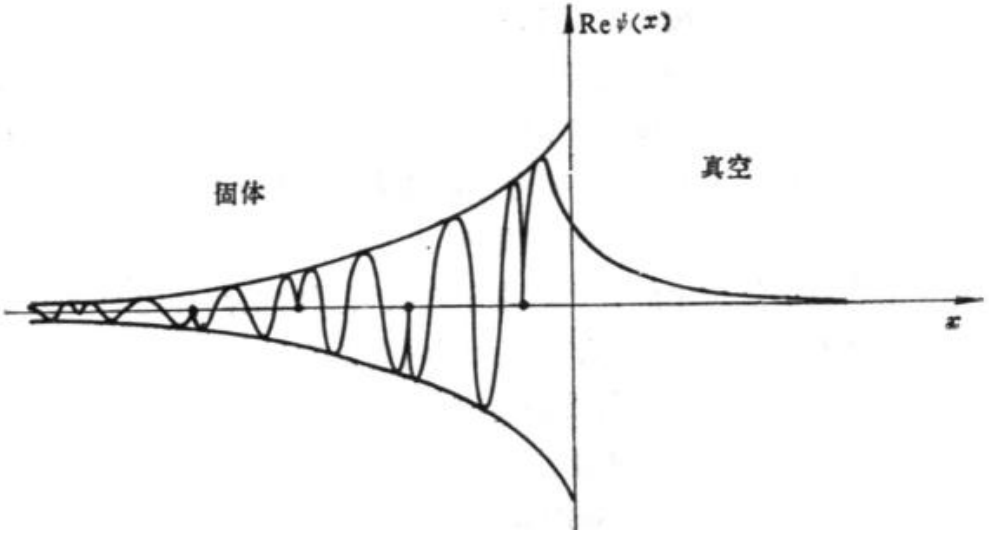
$$\psi = ae^{-\alpha z}$$

- 可确定q、a/c 值
- 再将q值带回能量表达式，即可得到相应的能量

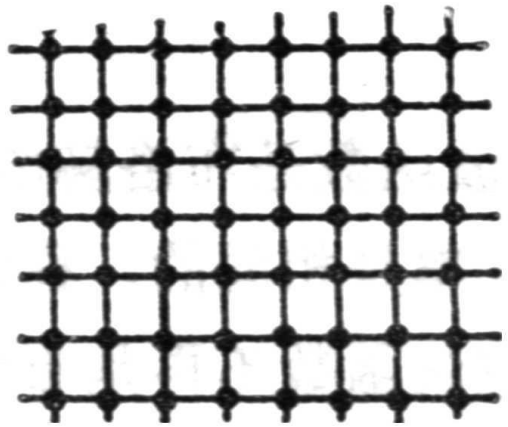
半无限长晶体：

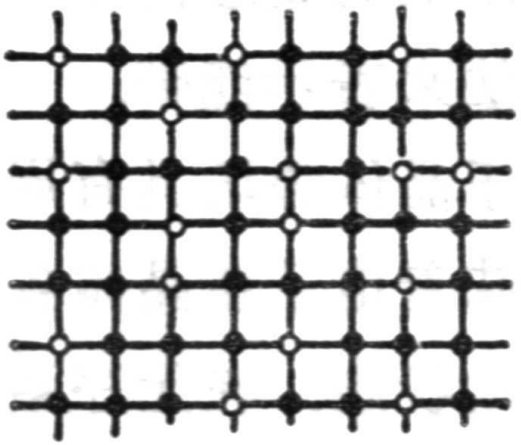
结论：

- 当k 为实数，晶体内部能带与一般晶体的情况一样
- 当k 为虚数，波函数在晶体内部是衰减的，能量本征值位于能隙之中，在这些解中有一个可以与真空区域的波函数相匹配，表明在表面内附近很窄的区域中有一个电子态，称为**表面态**。界面内外的波函数如图所示。

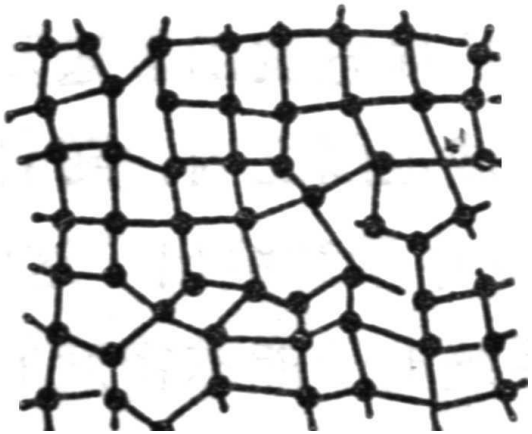


4.9 无序系统中的电子态









组分无序
位置无序
拓扑无序



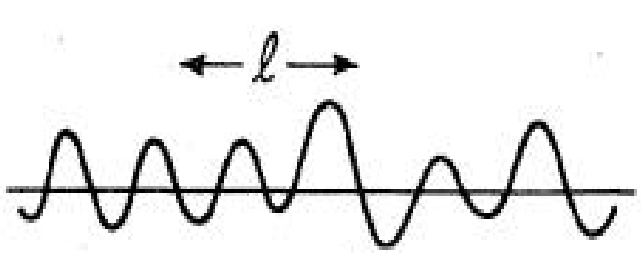
P.W. Anderson（Nobel Prize in Physics, 1977）

Anderson Localization (Phys Rev **109,**1492(1958))

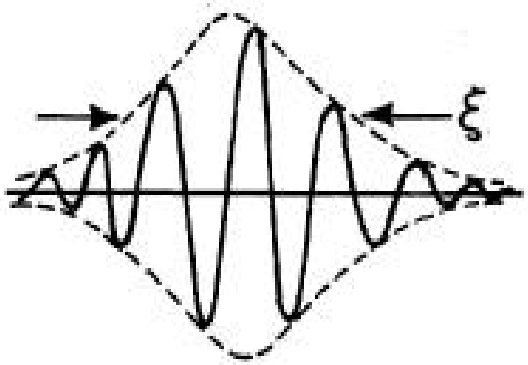
In 1958, Anderson pointed out that the electric wave function in a random potential may be profoundly altered if the randomness is sufficiently strong. The traditional view had been that scattering by the random potential causes the Bloch waves to lose phase coherence on the length scale of the mean free path l. Nevertheless, the wave function remains extended throughout the sample. Anderson pointed out that if the disorder is very strong, the wave function may become localized, in that the envelope of the wave function decays exponentially from some point in space,

$$|\psi(\mathbf{r})| \sim \exp(|\mathbf{r}-\mathbf{r}_0|/\xi)$$

Anderson 提出，当无序程度很强时，波函数会发生局域化。



扩展态的波函数



局域化的波函数

$$H=\sum_i \varepsilon_i |i\rangle\langle i| + \sum_{ij} V_{ij} |i\rangle\langle j|$$

Anderson **模型**

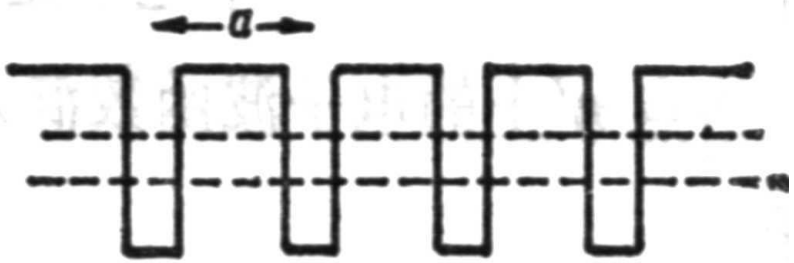
对角无序

$$H=\sum_i \varepsilon_i |i\rangle\langle i| + \sum_{ij} V_{ij} |i\rangle\langle j|$$

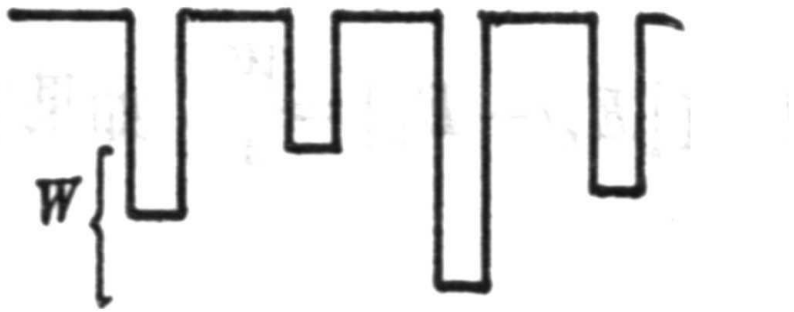
$$P(\varepsilon_i)=\frac{1}{W}, \left|\varepsilon_i-\bar{\varepsilon}\right|<\frac{W}{2}$$

$$a_n(t=0)=1$$

$$a_n(t\to\infty)=\begin{cases} 0 & extended \\ finite & localized \end{cases}$$



规则势场：周期性势阱模型



无序势场：安德森无序势阱模型

W 体现无序强度

$$H=\sum_i \varepsilon_i |i\rangle\langle i| + \sum_{ij} V_{ij} |i\rangle\langle j|$$

有序时，带宽 $B, B=2zV$ z配位数

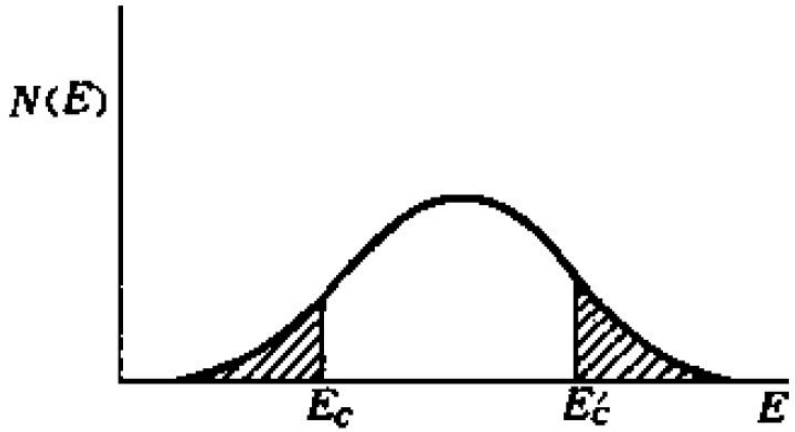
$W \gg B$时，

$$\psi = |n\rangle + \sum_{n'} \frac{V}{\varepsilon_{n'} - \varepsilon_n} |n'\rangle + \sum_{n''n'} \frac{V^2}{\left(\varepsilon_{n'} - \varepsilon_n\right)\left(\varepsilon_{n''} - \varepsilon_{n'}\right)} |n''\rangle + \cdots$$

收敛，波函数局域化；

$W \ll B$ 时，扩展态。

$|n\rangle$ 在能带尾部时，容易局域化；在能带中间容易延展。



1. 在确定能量，一个态只能要么是局域态，要么是延展态；
2. 波矢没有定义，Bloch态不存在，能带存在。无色散关系，电子结构用态密度表示；
3. 存在迁徙率边（Mobility Edge），用于区分局域态或扩展态。



Nevill F. Mott 1905-1996

Nevill F Mott (1905-1996)
Nobel Prize in Physics 1977

莫特提出了Mobility Edge（迁移率边）和Metal-Insulator Transition (MIT)

Anderson转变：无序导致的MIT,
Mott转变：多电子作用（依赖于电子浓度）导致的MIT

Proc. Phys. Soc. 1937, 49, 72
Rev. Mod. Phys. **1968**, 40, 677
Proc. R. Soc. Lond. A **1982**, 382, 1



P. W. Anderson, Phys. Rev. 109, 1492 (1958)
Phys. Rev. Lett. **1979**, 42, 673

Phys. Rev. Lett. 1980, 45, 1723
Phys. Rev. B 1983, 27, 7509
(P-doped Si)

▪ Hamiltonian and Methodology

• **2D** repulsive Hubbard Hamiltonian

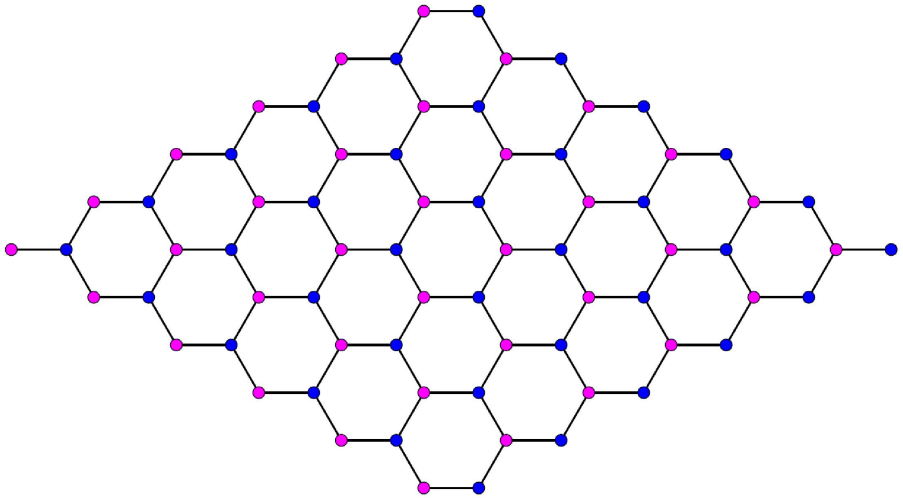
$$H = -\sum_{\langle \mathbf{ij}\rangle\sigma}\left(t_{\mathbf{ij}}c^{\dagger}_{\mathbf{i}\sigma}c_{\mathbf{j}\sigma}+\mathrm{h.c.}\right)-\mu\sum_{\mathbf{i}\sigma}n_{\mathbf{i}\sigma}+U\sum_{\mathbf{i}}\left(n_{\mathbf{i}\uparrow}-\frac{1}{2}\right)\left(n_{\mathbf{i}\downarrow}-\frac{1}{2}\right).$$

• hopping $t_{ij} \in [1-\Delta/2, 1+\Delta/2]$ ⇒ Bond disorder

• $\mu = 0$ **half-filled state**, ⇒ particle-hole symmetric **remained**

⇓

Sign problem free



• Sketch for L=6

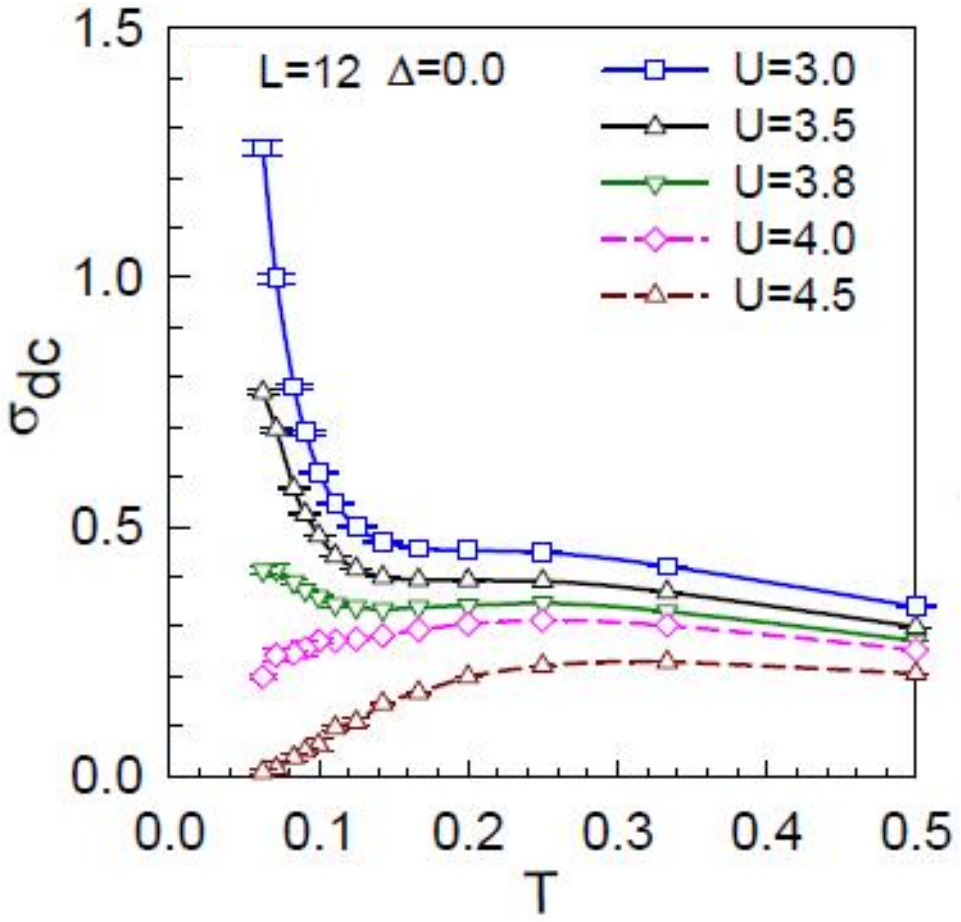
• The lattice geometries for a $2 \times L^2$ honeycomb lattice with periodic boundary conditions. $L = 3, 6, 9, 12, 15$

- **Results**

❑ Metal-insulator transition

Phys. Rev. Lett. **120**, 116601 (201

- temperature-dependent DC conductivity ➡ $\sigma_{\rm dc}$



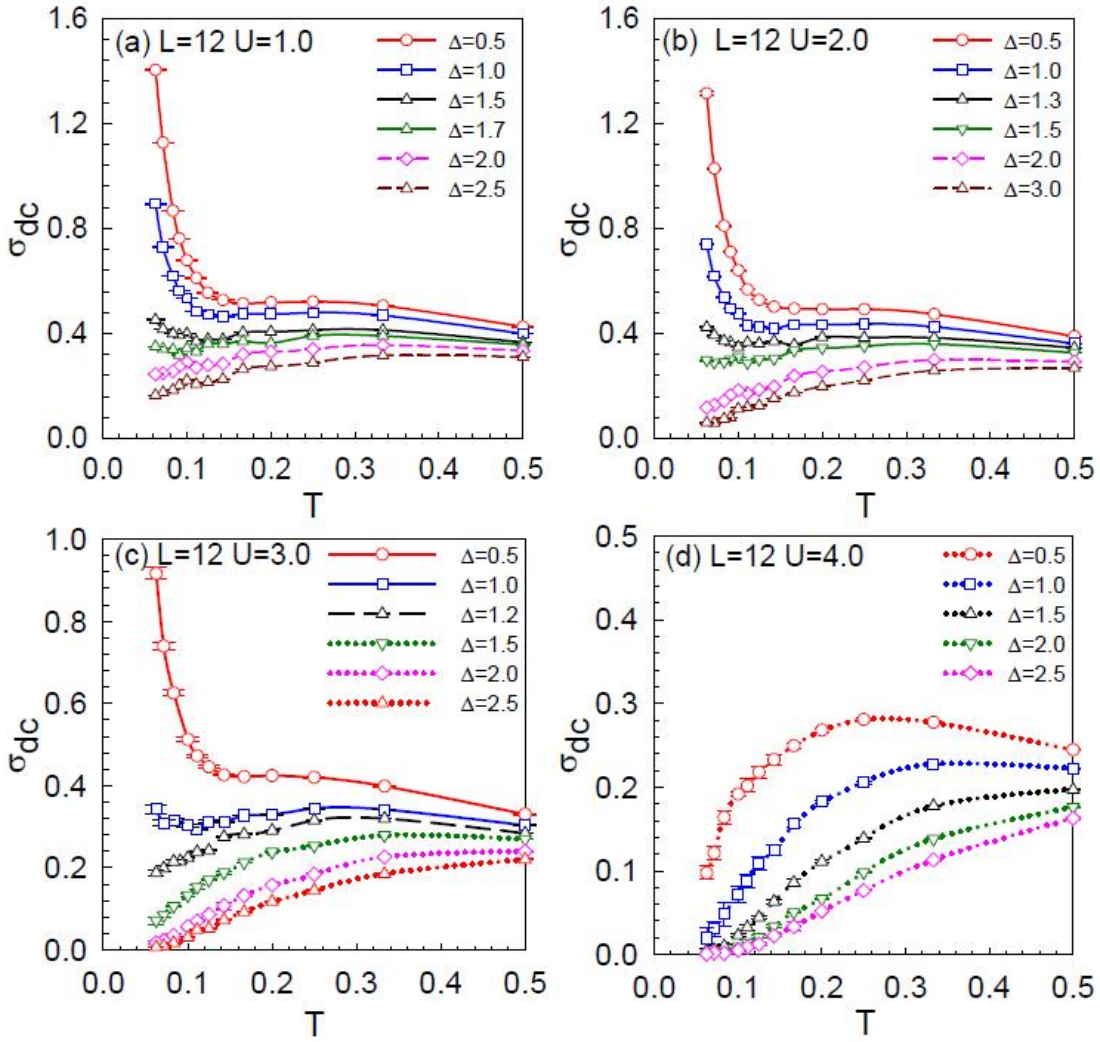
$$\sigma_{\rm dc} = \frac{\beta^2}{\pi}\Lambda_{xx}(\mathbf{q}=0, \tau=\beta/2)$$

Phys. Rev. Lett. 83, 4610 (1999).
Phys. Rev. Lett. 75, 312(1995).
Phys. Rev. B 54, R3756 (1996)

- $T \to 0 \;\; U \lesssim 3.8 \;\; d\sigma_{\rm dc}/dT < 0$ metallic
- $T \to 0 \;\; U \gtrsim 4.0 \;\; d\sigma_{\rm dc}/dT > 0$ insulating
- Critical point $U_c \sim 3.85$

Y. Otsuka, S. Yunoki, and S. Sorella, Phys. Rev. X 6, 011029 (2016).

❑ Temperature dependence of DC conductivity



- **bond-disorder** introduces a phase transition

- (a) $U = 1.0\ \Delta = 1.7$
- (b) $U = 2.0\ \Delta = 1.5$
- (c) $U = 3.0\ \Delta = 1.0$
- (d) $U = 4.0\ \Delta = 0.0$

Metallic

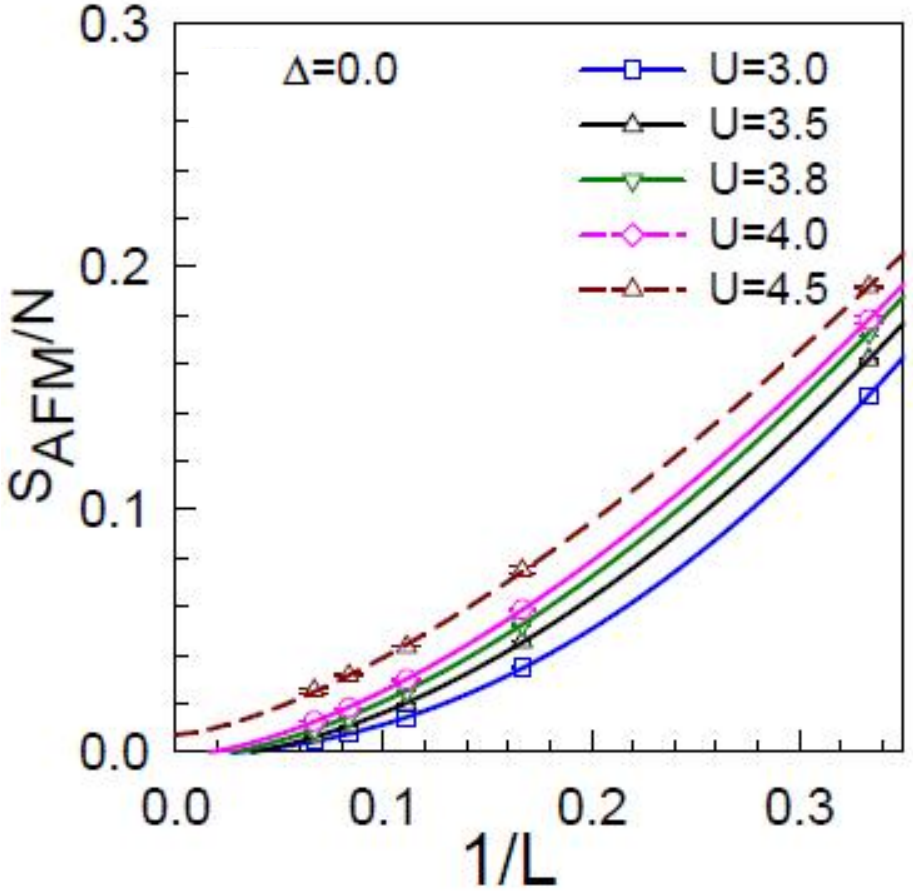
↓

Mott-Slater insulator

❑ Magnetic properties

- Antiferromagnetic(AF) structure factor at $Q = \Gamma$

$$S_{AF} = \frac{1}{N_c} \left\langle \left\langle \left(\sum_{\mathbf{r}\in A} \hat{S}^z_{\mathbf{r}} - \sum_{\mathbf{r}\in B} \hat{S}^z_{\mathbf{r}} \right)^2 \right\rangle \right\rangle_{\Delta} \xrightarrow{T \to 0} S_{AF} = \frac{M^2}{3} + \frac{a}{L}$$

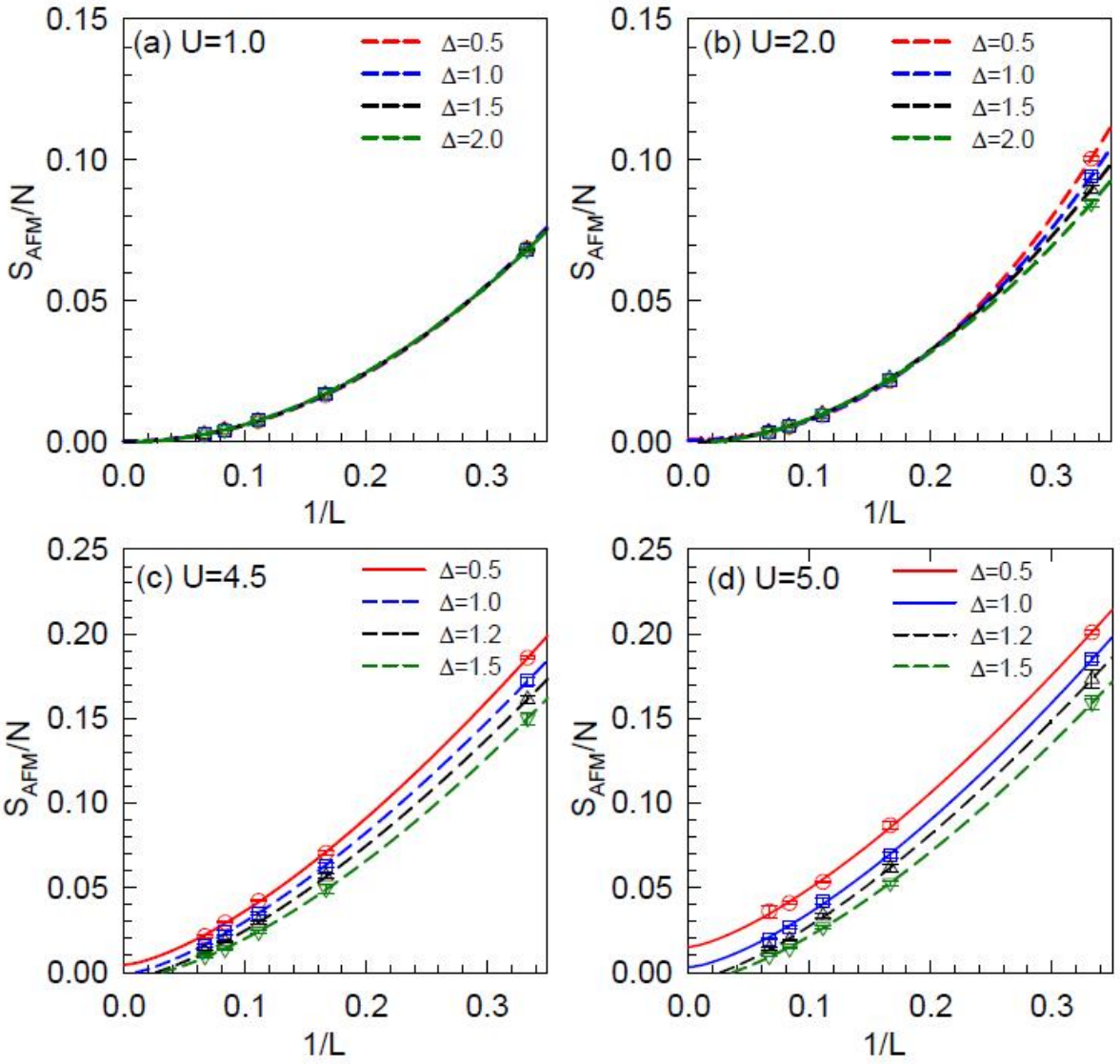


- Finite-size scaling studies

- Critical point $U_c \sim 3.85$

Y. Otsuka, S. Yunoki, and S. Sorella, Phys. Rev. X 6, 011029 (2016).

❑ Magnetic properties



- **bond-disorder** introduce a phase transition

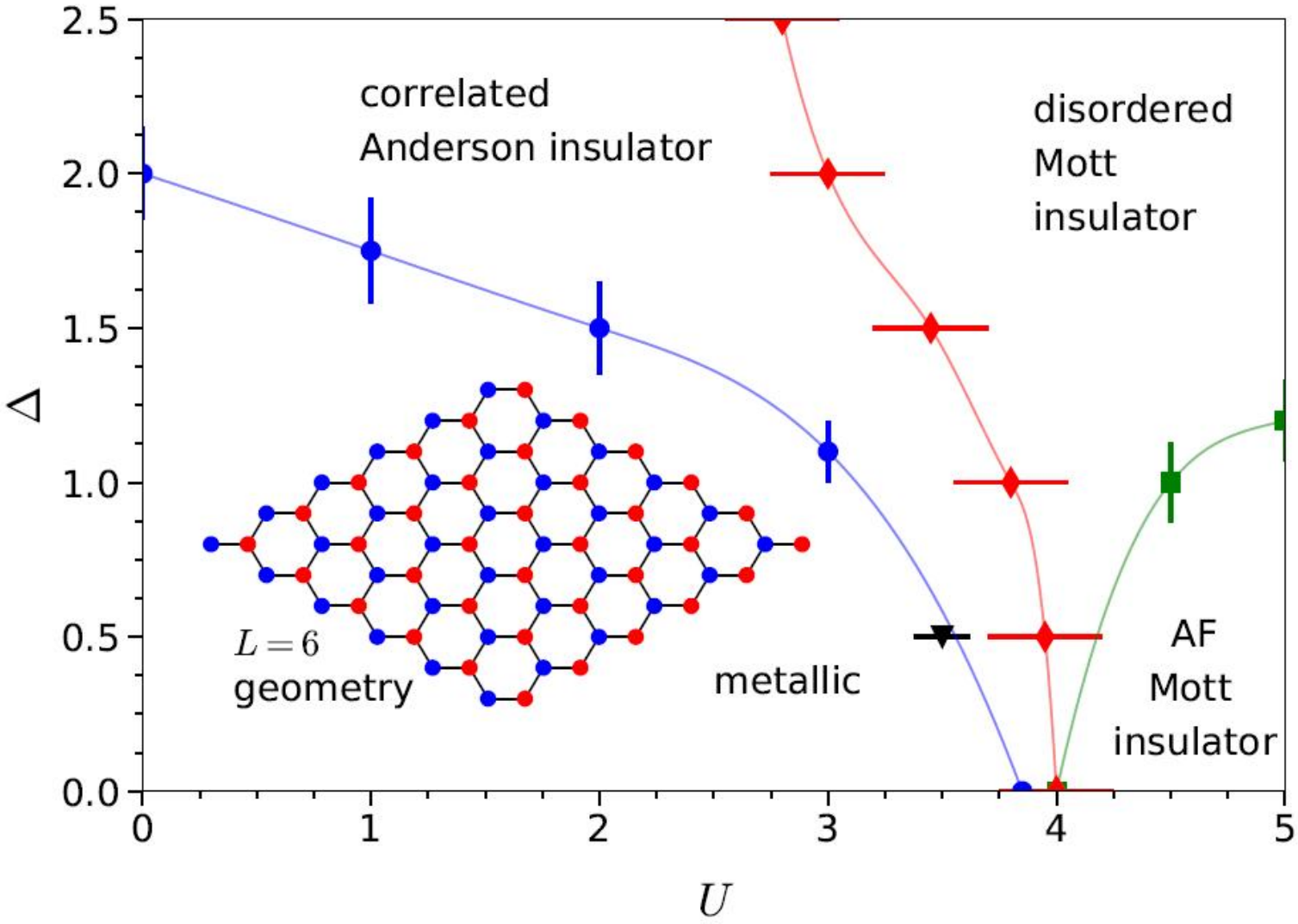
- (a) $U = 1.0$ no AF order
- (b) $U = 2.0$ no AF order
- (c) $U = 4.5\ \Delta = 1.0$
- (d) $U = 5.0\ \Delta = 1.2$

AF order

↓

Non-AF order

❑ Phase diagram

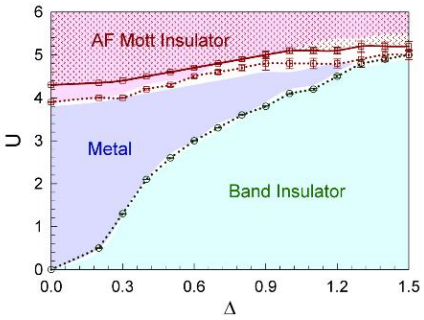


staggered potential

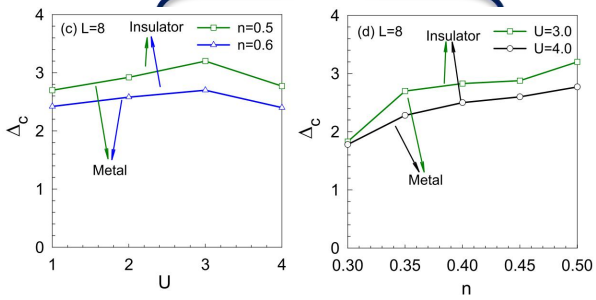
Strongly interacting model

interaction

doping



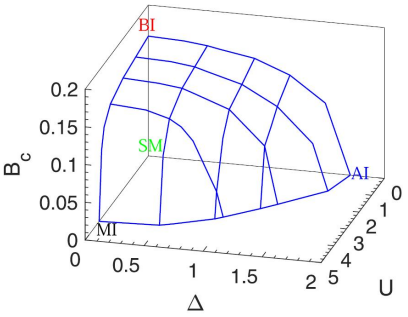
PRB 101, 245161 (2020)

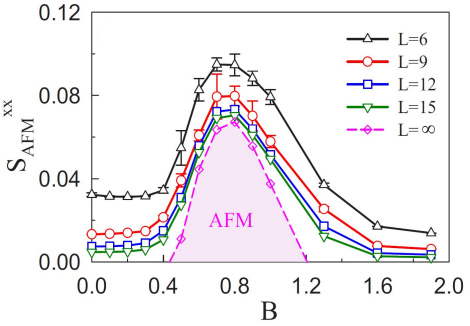


PRB 105, 045132(2022)
PRB 106, 205149(2022)

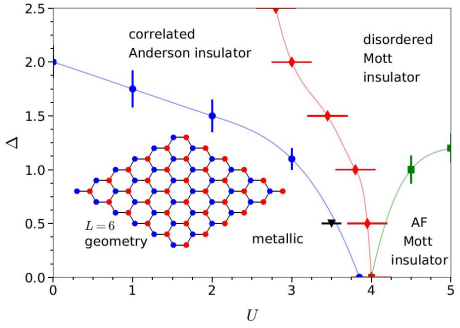
Magnetic field

disorder

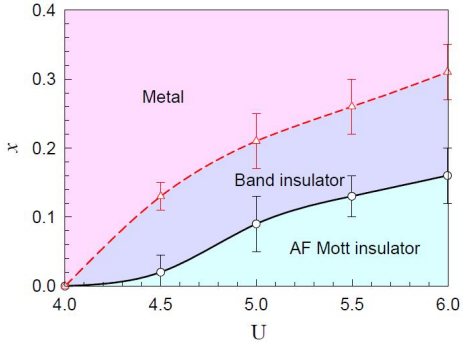




PRB 104, 045138 (2021) PRB 105, 045138 (2022)



PRL 120, 116601 (2018)



PRB 106, 205144(2022)